```
let lang='en';const T=(e,a)=>lang==='ar'?a:e;const $=i=>document.getElementById(i
const prof=[.2,.15,.12,.12,.15,.25,.5,.85,1,.8,.6,.55,.6,.65,.6,.65,.8,1,.95,.8,.
const CW={motorway:4,trunk:3,primary:2,secondary:1},CC={motorway:'#c2410c',trunk:
const EP=['https://overpass-api.de/api/interpreter','https://overpass.kumi.system
const S={tab:'explore',hour:8,mode:'off',hs:[],sel:null,hi:0,gain:[10,10,10,10],r
const map=L.map('map',{zoomControl:true}).setView([21.5433,39.1728],12);
const tl=L.tileLayer('https://tile.openstreetmap.org/{z}/{x}/{y}.png',{maxZoom:19
let tOk=0,tBad=0;
function mapMsg(h){const m=$('mapmsg');if(!h){m.style.display='none';return}m.inn
tl.on('tileload',()=>{tOk++;if(tOk>=2)mapMsg('')});
tl.on('tileerror',()=>{tBad++;if(tOk===0&&tBad>=4){S.tileErr=true;mapMsg(`<b>${T(
window.addEventListener('online',()=>{tBad=0;tl.redraw()});
const roadL=L.layerGroup().addTo(map),ixL=L.layerGroup().addTo(map),hsL=L.layerGr
const OPT=[
{en:'Signal-timing change',ar:'تعديل توقيت الإشارات',get e(){return T('Low','منخفض'
{en:'Intersection redesign',ar:'إعادة تصميم التقاطع',get e(){return T('High','تفع
{en:'Additional lane',ar:'مسار إضافي',get e(){return T('Medium','متوسط')},get d()
{en:'Proposed connecting road',ar:'طريق ربط مقترح',get e(){return T('High','مرتفع
const hash=id=>{const x=Math.sin((id%100003)*12.9898)*43758.5453;return x-Math.fl
const val=(c,h=S.hour)=>Math.min(100,Math.round(c.w*14*(.4+.6*c.h)*prof[h]));
const colr=v=>v<40?'#1f9d68':v<70?'#e8a20c':'#d63b3b';
const nmOf=n=>n||T('unnamed road','طريق بدون اسم');

async function loadRoads(){
 if(map.getZoom()<13){S.stat=T('Zoom in to level 13 or closer, then tap Load road
 S.stat=T('Loading real roads from OpenStreetMap…','جارٍ تحميل الطرق الحقيقية من O
 const b=map.getBounds(),q=`[out:json][timeout:25];way["highway"~"^(motorway|trun
 for(const u of EP){try{const r=await fetch(u,{method:'POST',body:'data='+encodeU
 S.stat=T('Could not reach the OpenStreetMap road-data service (Overpass). Check
function ingest(els){
 roadL.clearLayers();ixL.clearLayers();const nodes=new Map();let nw=0;
 els.filter(e=>e.type==='way'&&e.geometry).forEach(w=>{nw++;const t=w.tags||{},hw
  const pl=L.polyline(ll,{color:CC[hw]||'#888',weight:CT[hw]||2,opacity:.85}).add
  const lab=nmOf(t.name||t['name:en']||t.ref);
  pl.on('click',ev=>{L.DomEvent.stopPropagation(ev);S.sel={ll:[ev.latlng.lat,ev.l
   L.popup().setLatLng(ev.latlng).setContent(`<b>${lab}</b><br>OSM: ${hw}${t.ref?
  (w.nodes||[]).forEach((id,i)=>{let n=nodes.get(id);if(!n){n={id,ll:ll[i],ways:n
 S.cand=[...nodes.values()].filter(n=>n.ways.size>=2).map(n=>{const ws=[...n.ways
  return{id:n.id,ll:n.ll,w:ws.reduce((a,x)=>a+(CW[x.hw]||1),0),h:hash(n.id),names
 S.cand.sort((a,b)=>b.w-a.w).slice(0,500).forEach(c=>L.circleMarker(c.ll,{radius:
 S.stat=T(`Loaded ${nw} real road segments and ${S.cand.length} intersections (Op
 hotspots();render()}
function hotspots(){
```

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 hsL.clearLayers();const c=S.cand.map(x=>({...x,v:val(x)})).sort((a,b)=>b.v-a.v),
 for(const x of c){if(out.length>=8)break;if(out.every(o=>map.distance(o.ll,x.ll)
 S.hs=out;out.forEach((x,i)=>{x.name=x.names.slice(0,2).join(' × ')||T('Unnamed j
  L.marker(x.ll,{icon:L.divIcon({className:'',html:`<div class="hs" style="backgr
 if(S.hi>=out.length)S.hi=0}
map.on('click',e=>{if(S.mode==='new'){S.cur.push([e.latlng.lat,e.latlng.lng]);dra
function drawCur(){curL.clearLayers();if(S.cur.length){L.polyline(S.cur,{color:'#
function drawPr(){prL.clearLayers();S.roads.forEach((r,i)=>{L.polyline(r.pts,{col
 S.imps.forEach(m=>L.marker(m.ll).addTo(prL).bindTooltip(m.opt))}
const len=p=>p.reduce((a,x,i)=>i?a+map.distance(p[i-1],x):0,0);
function near(pts){if(pts.length<2)return[];const s=[];for(let i=1;i<pts.length;i
 return S.hs.map((h,i)=>({i,h,d:Math.min(...s.map(p=>map.distance(p,h.ll)))})).fi
const km=m=>(m/1000).toFixed(2)+' km';
const hsel=()=>S.hs.length?`<select onchange="S.hi=+this.value;map.setView(S.hs[S
const sim=()=>`<span class="chip">${T('SIMULATED','محاكاة')}</span>`;
const hourUI=()=>`<label>${T('Hour (demo daily pattern)','لساعة (نمط يومي تجريبي)

const V={
explore(){return `<div class="card"><h3>${T('Explore real roads','الطرق الحقيقية
 <button class="b" onclick="loadRoads()">${T('Load roads in view','الطرق في العرض
 <p class="mut">${S.stat}</p>${S.sel?`<p><b>${T('Selected','المحدد')}:</b> ${S.se
 <div class="card"><h3>${T('Simulated congestion hotspots','نقاط ازدحام محاكاة')}
 ${S.hs.map((h,i)=>`<div class="row"><span>#${i+1} ${h.name}</span><b style="colo
propose(){const r=S.roads.map((x,i)=>`<div class="row"><span>${T('Proposed road',
 return `<div class="card"><h3>${T('Propose a new road','اقتراح طريق جديد')}</h3>
 <button class="b" onclick="S.mode=S.mode==='new'?'off':'new';render()">${S.mode=
 <button class="b alt" onclick="if(S.cur.length>1){S.roads.push({pts:S.cur,desc:S
 <p class="mut">${T('Current points','النقاط الحالية')}: ${S.cur.length}${S.cur.l
 <label>${T('Describe the proposal','صف الاقتراح')}</label><textarea onchange="S.n
 <div class="card"><h3>${T('Propose an improvement','اقتراح تحسين')}</h3><p class
 <select id="io">${OPT.map(o=>`<option>${T(o.en,o.ar)}</option>`).join('')}</sele
 <button class="b" onclick="if(S.sel){S.imps.push({ll:S.sel.ll,label:S.sel.label,
 ${S.imps.map(m=>`<div class="row"><span>${m.opt}</span><span class="mut">${m.lab
compare(){if(!S.hs.length)return `<div class="card"><h3>${T('Planning dashboard',
 return `<div class="card"><h3>${T('Compare solutions','قارن الحلول')} ${sim()}</
 <p class="mut">${T('Crossing roads (OSM)','الطرق المتقاطعة (OSM)')}: ${h.names.j
 <table><tr><th>${T('Option','الخيار')}</th><th>${T('Effort','الجهد')}</th><th>${
 ${OPT.map((o,i)=>`<tr><td>${T(o.en,o.ar)}<br><span class="mut">👍 ${T(o.p[0],o.p
 <p class="mut">⚠️ ${T('What-if gains are YOUR assumptions (same placeholder defa
 ${S.roads.map((r,i)=>{const n=near(r.pts);return `<div class="card"><h3>${T('Pro
 <p class="mut">${T('Drawbacks to study: land acquisition, environment, cost, new
report(){const h=S.hs[S.hi];if(!h)return `<div class="card">${T('Load roads first
 const t=[T('JEDDAH TRAFFICFLOW AI – DRAFT PLANNING SUMMARY','ملخص تخطيطي مبدئي –
 T('1. Location','1. الموقع')+': '+h.name+' ('+h.ll[0].toFixed(5)+', '+h.ll[1].to
 T('2. Simulated score','2. الدرجة المحاكاة')+': '+h.v+'/100 @ '+String(S.hour).p
 T('3. Road classes at junction (OSM)','3. فئات الطرق عند التقاطع (OSM)')+': '+h.
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 T('4. Options considered (what-if, user assumptions)','4. (افتراضات المستخدم) سة
 T('5. Proposed new roads','5. طرق جديدة مقترحة')+': '+(S.roads.length?S.roads.ma
 T('6. Proposed improvements','6. تحسينات مقترحة')+': '+(S.imps.length?S.imps.map
 T('7. Before any decision: collect real counts/speeds/signal data, validate with
 return `<div class="card"><h3>${T('Report','التقرير')} ${sim()}</h3>${hsel()}<di
};
const TABS=[['explore','Explore','استكشاف'],['propose','Propose','اقتراح'],['comp
function render(){document.documentElement.lang=lang;document.documentElement.dir
 $('ttl').textContent=T('Jeddah TrafficFlow AI','جدة ترافيك فلو AI');$('sub').tex
 $('wn').innerHTML='<b>'+T('Real:','حقيقي:')+'</b> '+T('road geometry from OpenSt
 $('nv').innerHTML=TABS.map(t=>`<button class="${S.tab===t[0]?'on':''}" onclick="
 const y=window.scrollY;$('panel').innerHTML=V[S.tab]();window.scrollTo(0,y)}
$('lg').onclick=()=>{lang=lang==='ar'?'en':'ar';hotspots();render()};
map.on('locationerror',()=>{S.stat=T('Location unavailable. Allow location for Sa
render();
```